- For a mixed or entangled state:

$$\rho^2 \neq \rho$$

$$Tr(\rho^2) < 1$$

The last two properties give us a clear way to distinguish mathematically between pure and mixed states. A subsystem of an entangled state (such as Alice's half of the singlet state) is considered a mixed state.

It's worth taking a moment to understand these two properties a little better. To simplify things, we will assume that ρ is a diagonal matrix—in other words, all of its off-diagonal elements are zero. This simplification costs us nothing because ρ is Hermitian, and it turns out that every Hermitian matrix can be expressed in diagonal form in some basis.[7] Taking the square of a diagonal matrix is quite simple: all you need to do is square each individual element. Since ρ represents a mixed state, and the diagonal elements of ρ must add up to 1, *none* of the diagonal elements of ρ can *equal* 1. Otherwise, ρ would represent a pure state. Therefore, ρ must have at least two positive diagonal elements that are less than 1. Squaring these elements gives a new matrix ρ^2 whose elements are even smaller. This accounts for both of the mixed-state properties of ρ.

Before you try the next exercises, I'll mention one more thing about the trace. It turns out that the trace has many interesting mathematical properties. One of its more useful properties is that the trace of a product of two matrices does not depend on their order of multiplication. In other words,

$$TrAB = TrBA,$$

even if